Smart Job Matching: How Recommender Systems Work on LinkedIn & Indeed

AI-POWERED CAREER DISCOVERY

[Presenter Name] • [Date] • [Company/Institution]

THE JOB SEARCH CRISIS

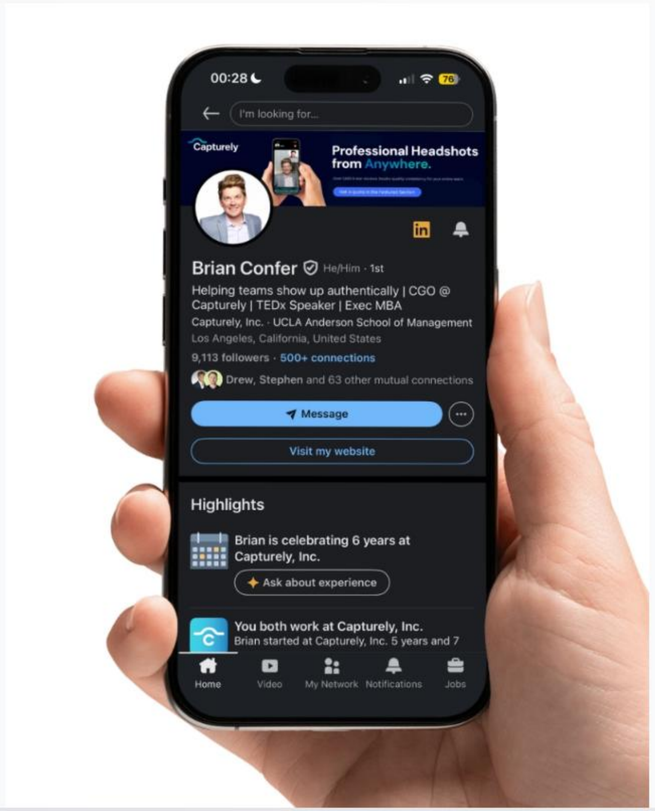
With millions of active listings, the traditional "search and scroll" method is no longer sustainable for job seekers or recruiters.

- **Information Overload:** Average user spends hours filtering irrelevant posts.
- **The Visibility Gap:** High-quality candidates often missed by standard keyword searches.
- **AI Intervention:** Recommenders solve the "Needle in a Haystack" problem by predicting relevance.

> *"How does LinkedIn know exactly which job to show you first? It's not magic—it's advanced data science."*



WHAT IS A JOB RECOMMENDER?



Dual-Directional Matching

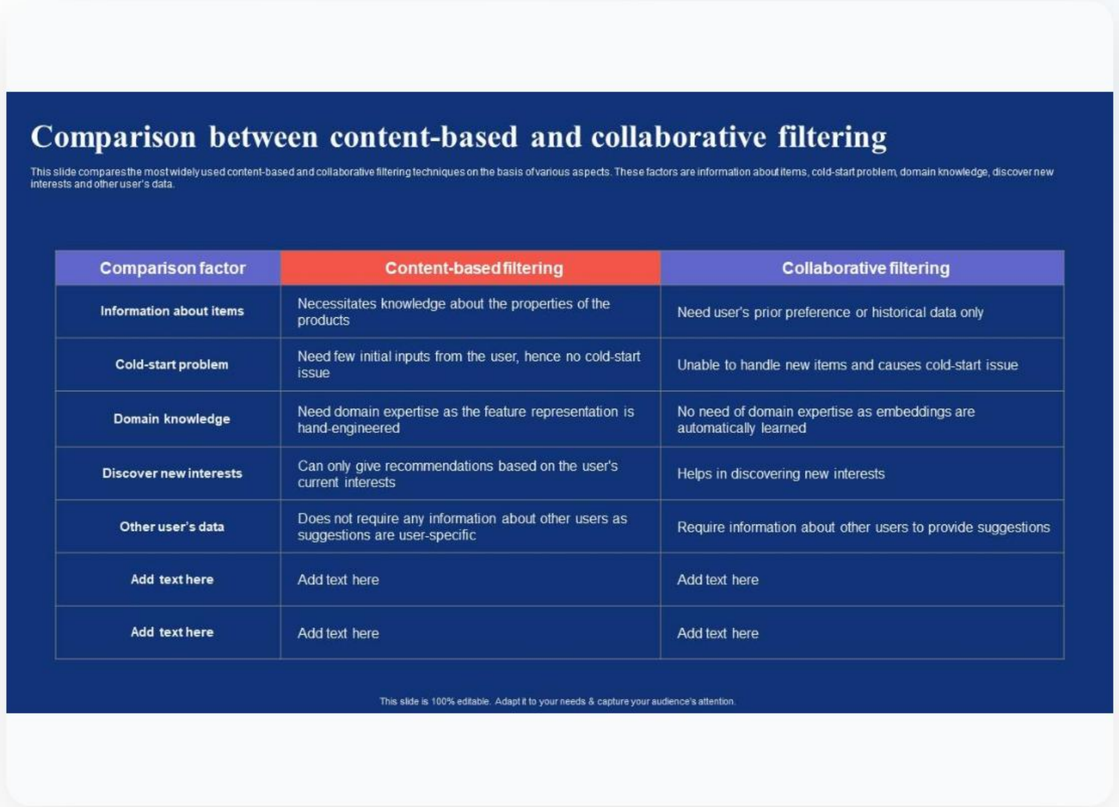
Unlike entertainment apps, job recommenders act as a **two-sided marketplace** system:

- **Job-to-Candidate:** Proposing listings to a user ("Jobs You May Like").
- **Candidate-to-Job:** Suggesting top profiles to recruiters.
- **UI Integration:** Embedded in feeds, emails, and the "Jobs" tab.

Goal: Maximize "Hire Probability" for every interaction.

RECOMMENDATION ARCHITECTURES

Approach	Mechanism	Strength
Content-Based	Matches keywords to user profile skills.	Works for niche skills; no popularity bias.
Collaborative	"Users like you applied to X."	Discovers hidden patterns; less data entry.
Hybrid Models	Combination of both via weighted stacking.	Highest accuracy; handles varied data.



Comparison between content-based and collaborative filtering

This slide compares the most widely used content-based and collaborative filtering techniques on the basis of various aspects. These factors are information about items, cold-start problem, domain knowledge, discover new interests and other user's data.

Comparison factor	Content-based filtering	Collaborative filtering
Information about items	Necessitates knowledge about the properties of the products	Need user's prior preference or historical data only
Cold-start problem	Need few initial inputs from the user, hence no cold-start issue	Unable to handle new items and causes cold-start issue
Domain knowledge	Need domain expertise as the feature representation is hand-engineered	No need of domain expertise as embeddings are automatically learned
Discover new interests	Can only give recommendations based on the user's current interests	Helps in discovering new interests
Other user's data	Does not require any information about other users as suggestions are user-specific	Require information about other users to provide suggestions
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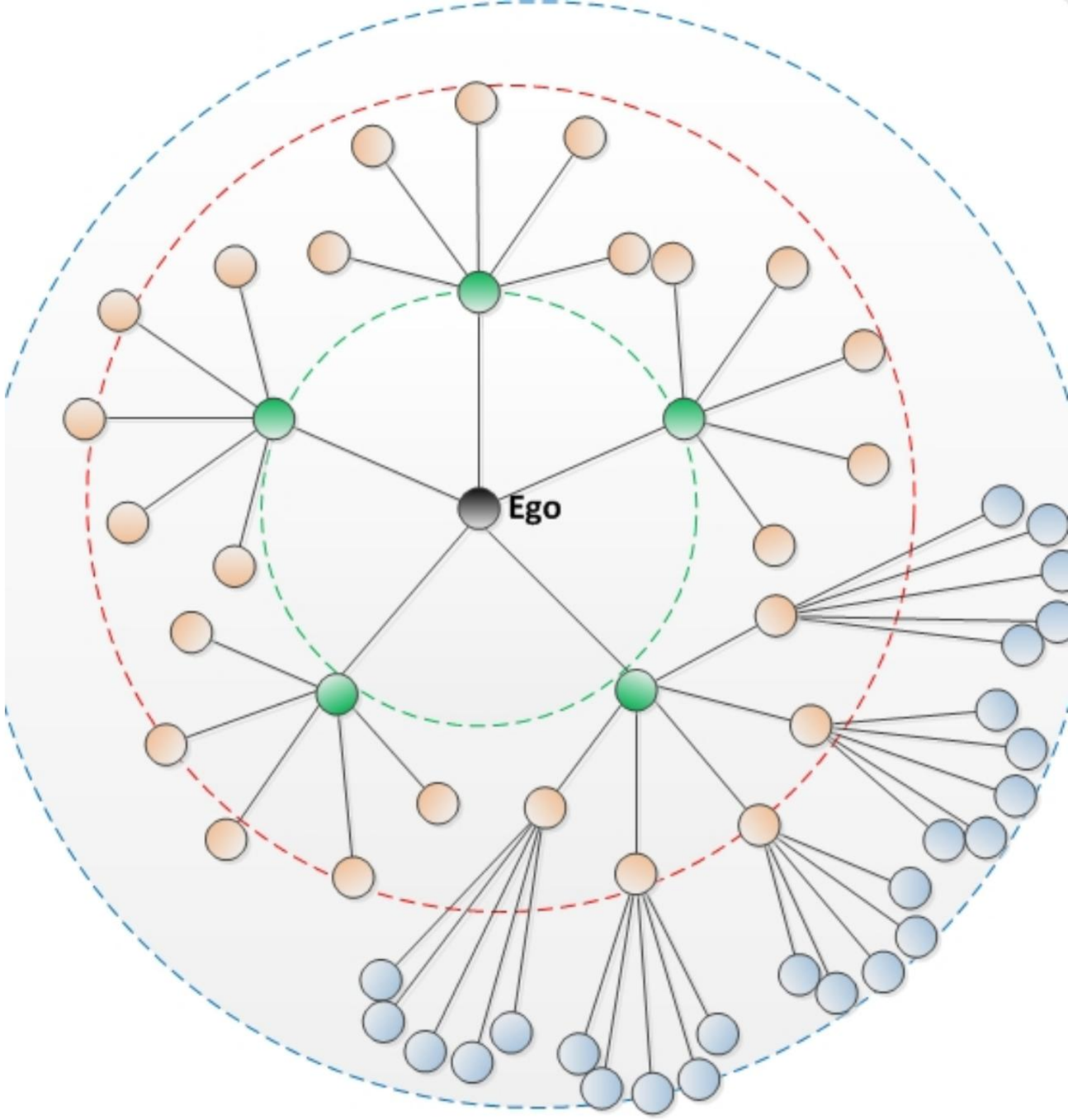
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LINKEDIN: THE SOCIAL GRAPH

The JUDE Framework

LinkedIn utilizes a **Two-Tower Neural Network** to generate high-dimensional embeddings for members and jobs.

- **Social Signals:** Connections at target companies increase relevance.
- **GNN Analysis:** Graphs determine how "close" a member is to a role.
- **Personalization:** Considers seniority and location preferences.



A Pure Branching Network (with Degrees of Separation)

INDEED: THE SEARCH POWERHOUSE



Intent-Driven Matching

Indeed focuses on **Resume Parsing** and **Real-time Intent** signals from search queries.

- **Skill Extraction:** Maps text to 40,000+ canonicalized skills.
- **Search Loops:** Clicks and skips refine models in milliseconds.
- **AI Coach:** LLMs explain why a job fits your resume.

CORE TECHNOLOGY STACK

NLP & BERT

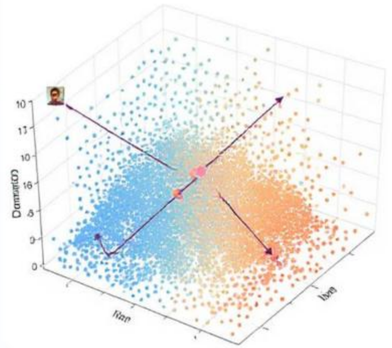
Understanding semantic context of titles like "Customer Success Ninja" vs "Support Rep".

Matrix Factorization

Predicting application probability by breaking down latent factors of user behavior.

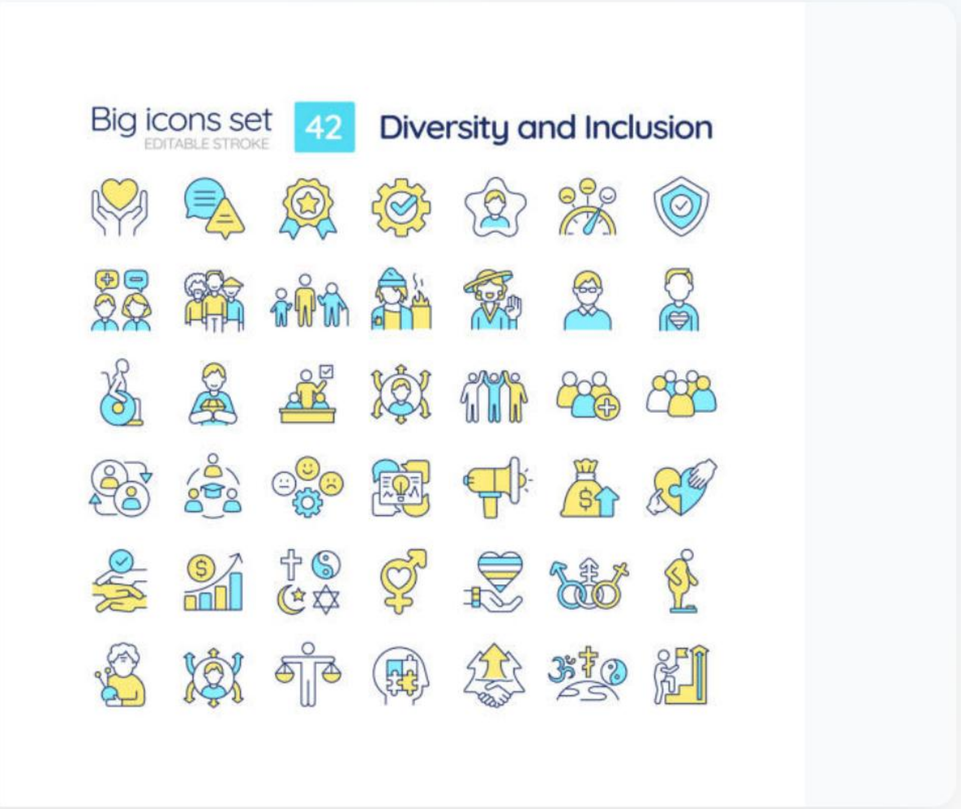
Learning-to-Rank

Final re-ranking using GBDT models to ensure top results match immediate needs.



ALGORITHMIC HURDLES

- **Cold Start:** Recommending for new users with zero history.
- **Bias & Fairness:** Preventing algorithmic perpetuation of historical bias.
- **Filter Bubbles:** Allowing for career pivots beyond past history.
- **Data Sparsity:** Users interact with very few jobs overall.



MEASURING SUCCESS



*Success is defined by both **Implicit** (clicks) and **Explicit** (hires) signals.*

THE LLM REVOLUTION



Explainability

- **Generative Recs:** "You are seeing this because your Python skills match the requirement."
- **Multi-Tasking:** Predicting salary fit and commute alongside skill match.
- **Semantic Understanding:** LLMs reduce mismatch by 40% over keywords.

LINKEDIN VS. INDEED

Feature	LinkedIn	Indeed
Data Core	Social Graph + Profile	Resume + Search Behavior
Advantage	Network insights	Massive search volume
Recruiter View	InMail & Branding	Smart Sourcing Suites



THE ROAD AHEAD

Conversational Hiring

Moving from search bars to active AI career agents.

- **Skills-First:** Matching capabilities over formal degrees.
- **Privacy First:** Federated learning for secure personalization.
- **Real-time Market:** Economic trends driving suggestions.



Questions & Discussion

Recommender systems have transformed the labor market from a chaotic notice board into a surgical matching engine.



Your Smart Job Matching Journey Starts Here.

IMAGE SOURCES



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Source: medium.com

IMAGE SOURCES



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Source: www.istockphoto.com



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Source: www.androidpolice.com



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Source: www.hyperclapper.com



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